Floor Plan Studio

Project summary - rule-based house floor plan generator with CAD-style editing (Windows desktop app)

Purpose	Desktop tool that generates house floor plans from rules and lets the user edit them like a light AutoCAD. Built for a land surveyor in Jordan who works with architects and interior designers.
Origin	Started from a Blender Geometry Nodes tool (random grid subdivision). Rebuilt as a standalone app, no Blender dependency.
Repository	github.com/essamnawafleh-oss/floorplan-studio
Stack	Python 3.12, PySide6 (Qt) UI, shapely (geometry), ezdxf (CAD), trimesh (3D), numpy. PyInstaller + Inno Setup packaging, GitHub Actions CI, 53 automated tests.
Status	Version 0.3.0, working Windows installer and portable build published automatically on every push.

1. Generation engine

- **Inputs** - building size in 5 modes: fixed width x depth, fixed width (auto depth), fixed depth (auto width), plot + zoning (Amman Residential A/B/C/D setbacks and coverage), or automatic from areas.
- **Rooms** - either from the **household** (parents, sons, daughters, others, salon guests, office, maid) or a **custom room list** with fixed areas.
- **Space programme** - editable Jordanian defaults: boys and girls in separate rooms, max 2 per room, salon size from guest count, bathrooms from number of residents.
- **Algorithm** - guillotine slicing with a seeded random search (hundreds of candidates, best score wins); night wing with single or double-loaded corridor; small rooms clustered with their host (guest WC + entrance, laundry + kitchen); automatic footprint calibration so net areas match targets.
- **Seed** - changes the arrangement only; the footprint stays fixed. Same seed always gives the same plan.
- **Scoring rules** - minimum area and width, aspect ratio, window for every habitable room, reachability from the main door, no access through bedrooms, adjacency wishes (kitchen-dining, etc.). Violations become Arabic/English warnings.
- **Strict mode** - rejects any layout breaking a hard rule. Optional rooms can be removed (laundry, guest WC...).
- **Doors** - access tree from the main door. **Windows** - sized from the required glazing ratio.

2. User interface

- Arabic (RTL) and English; the plan updates live; 6 alternative thumbnails per input set.
- Rooms table (target vs actual area), warnings list.
- Themes: light, dark, midnight, sand, or custom JSON theme file; accent colour, UI font and size.
- Projects saved/opened as JSON, including manual edits.

3. Edit mode

- Drag interior and exterior walls (5 cm steps); set room size, type and name.
- Add, move and edit doors/windows: width, sill/head heights, swing direction, hinge side. Undo/redo.
- **Block library** - furniture, sanitary, kitchen, door and window blocks; DXF block import with unit detection; place and rotate furniture; swap door/window symbols.

- **Mini-AutoCAD** - command line with aliases L, PL, REC, C, A, T, DIM, O, TR, EX, X, M, CO, RO, SC, MI, E; input x,y / @dx,dy / @d<angle and direct distance entry; Enter/Space repeat, Esc cancel.
- Object snaps: endpoint, midpoint, center, intersection, perpendicular, quadrant, nearest (on drawing, walls and furniture); ORTHO (F8), GRID (F9), OSNAP (F3).
- Window/crossing selection, grips, layers and colours; quick-mode Trim, Extend, Offset, Explode (also explodes furniture blocks), Move, Copy, Rotate, Scale, Mirror, Erase.

4. Extra tools

- **Measure** - length and area with snapping; units m / cm / mm / ft.
- **Drawing scale** - live 1:N readout, scale bar, PDF at true scale (1:50, 1:100...) on A4-A1.
- **3D viewer** - built-in software renderer (no GPU needed): orbit, pan, zoom, hide furniture, lower walls to look inside; open the GLB in an external app.
- **Quantities** - blocks for interior/exterior walls (openings deducted); floor tiles per room and total (preset or custom sizes, joints, waste, boxes); paint for facades and interiors in litres and gallons; CSV export.
- **Python add-ons** (like Blender add-ons) - a .py file with register(api); built-in code editor with highlighting, enable/disable, 3 example add-ons (room schedule CSV, auto-furnish, wall/glazing summary).

5. Export formats

- **2D**: SVG, PNG, PDF.
- **CAD**: DXF with layers (walls, doors, glazing, room areas, text, dimensions); furniture as real block INSERTs; CAD drawing as real lines, arcs, text and dimensions.
- **3D**: GLB and OBJ. **Data**: JSON.

6. Distribution

- GitHub Actions: run tests, build on Windows, smoke-test the exe, publish **Setup.exe** and **portable zip** to Releases.
- Chrome "virus detected" false positive reduced by compiling the PyInstaller bootloader from source and shipping a one-folder build with an installer instead of a single self-extracting exe.
- The app is **not code-signed yet**. Planned fix: Microsoft Store (free, Store signs it) or a code-signing certificate.

7. Known limits / next steps

- Rule values and zoning presets are **approximate** and must be checked against current regulations.
- Rectangular plots and a single floor only.
- Automatic layouts can still produce some narrow small rooms; generator needs further improvement.
- CAD tools are separate from the generated walls. Not yet available: Fillet, Hatch, Array, Stretch, polar tracking.
- Planned: irregular plots from survey data, multiple floors with stairs, IFC export, Microsoft Store release, UI redesign from a mock-up image.